

Ch. 7 — P02 (A) M/M/c and the Erlang-C Formula

Problem Bank | Chapter 7 | Type: A | Difficulty:

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A hospital emergency department triage station receives patients at a rate of $\lambda = 12$ patients/hour following a Poisson process. Each triage nurse can assess a patient in an exponentially distributed time with mean $1/\mu = 10$ minutes ($\mu = 6$ /hour).

Tasks

1. **M/M/1 baseline:** Suppose $c = 1$ nurse is on duty. Is the system stable? Compute ρ , W_q (minutes), and L_q .
2. **M/M/c with $c = 2$:** Compute the offered load $a = \lambda/\mu$ and traffic intensity $\rho = \lambda/(c\mu)$. Using the Erlang-C formula:

$$C(c, a) = \frac{\frac{a^c}{c!} \cdot \frac{1}{1 - \rho}}{\sum_{n=0}^{c-1} \frac{a^n}{n!} + \frac{a^c}{c!} \cdot \frac{1}{1 - \rho}}$$

compute $C(2, 2)$ — the probability that an arriving patient must wait. Then compute W_q and L_q .

3. **M/M/c with $c = 3$:** Repeat Task 2 for $c = 3$ nurses.
4. **Comparison table:** Summarize ρ , $C(c, a)$, W_q (min), and L_q for $c \in \{1, 2, 3\}$ in a single table.
5. **Throughput check:** Using Little's Law ($L = \lambda W$), verify your L values for $c = 2$ and $c = 3$.
6. **Staffing decision:** The department's service standard is $W_q \leq 3$ minutes. What is the minimum number of nurses needed to meet this standard? Justify using your computed values.

Express W_q in minutes and all other quantities to three significant figures.